
Author

author: name: degrees: orcid: email: 1132236013@rudn.ru affiliation: - name:
country: postal-code: 117198 city: address: . -
, . 6 ## Title title: SIR subtitle: №6 license: CC
BY date: today date-format: "2026-04-25"

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- : -01-23
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- . .
- student@rudn.ru

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- : SIR (Susceptible–Infectious–Recovered)
- : SIR Julia

- SIR
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- : Julia
- : AlgebraicPetri, OrdinaryDiffEq, Plots, DataFrames, Random
- : (Tsit5)
- : (SSA)
- : Graphviz, Plots
- : Literate, Jupyter Notebook, Quarto

SIR

SIR

- : $S(\quad), I(\quad), R(\quad)$
- :
 - infection: $S + I \rightarrow I + I(\quad)$
 - recovery: $I \rightarrow R(\quad)$
- : $S=990, I=10, R=0$

-
- Tsit5
- : $S(t), I(t), R(t)$

-
- :
- a_inf = * S * I
- a_rec = * I
-
- :

1.

v.6.0.0, v.4.0.0

```
nothing to commit, working tree clean
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab06$ git flow release start
6.0.0
Switched to a new branch 'release/6.0.0'

Summary of actions:
- A new branch 'release/6.0.0' was created, based on 'develop'
- You are now on branch 'release/6.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '6.0.0'
```

1

2.

Julia

: AlgebraicPetri, OrdinaryDiffEq, Plots, DataFrames, Random

3.

sirpetri_run.jl

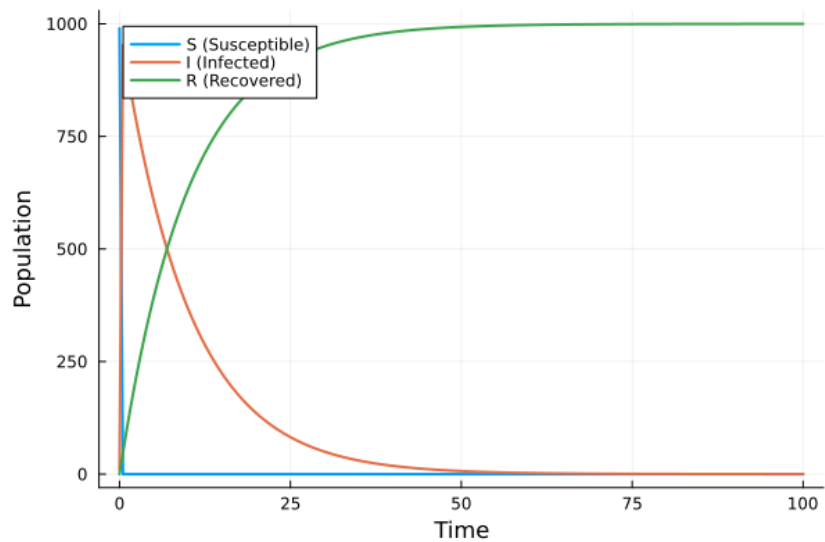
(=0.3, =0.1, tmax=100)

```
[e66e0078] + CompilerSupportLibraries_jll v1.1.1+0
[4536629a] + OpenBLAS_jll v0.3.23+4
[05823500] + OpenLibm_jll v0.8.1+2
[efcefd7] + PCRE2_jll v10.42.0+1
[bea87d4a] + SuiteSparse_jll v7.2.1+1
[8e850b90] + libblastrampoline_jll v5.11.0+0
Info Packages marked with ^ and x have new versions available. Those with ^ may be upgradable, but those with x are restricted by compatibility constraints from upgrading. To see why use `status --outdated -m`
Precompiling project...
183 dependencies successfully precompiled in 431 seconds. 284 already precompiled.
1 dependency had output during precompilation:
└─ StochasticDiffEqCore
   └─ WARNING: using DiffEqNoiseProcess.save_noise! in module StochasticDiffEqCore conflicts with an existing identifier.
└─
✓ Все пакеты установлены!
Для проверки: using DrWatson, DifferentialEquations, Plots
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_run.jl
Базовый прогон завершён. Результаты в data/ и plots/
```

2

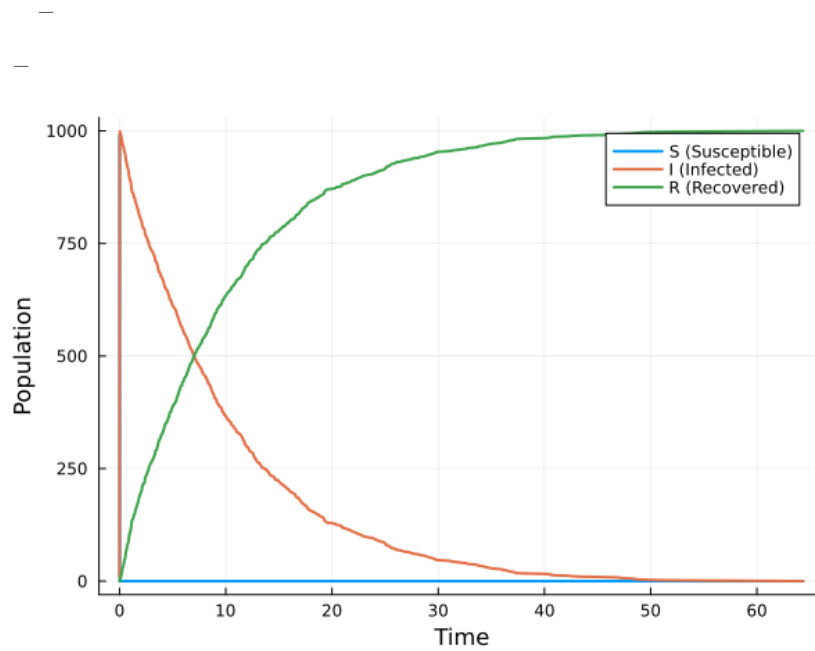
```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_run.jl
Базовый прогон завершён. Результаты в data/ и plots/
```

3



4

4



5

4. (sirpetri_scan_parameters.jl)

- 0.1 0.8 0.05
- I R

```

sirpetri_scan_parameters.jl
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/sirpetri_scan_parameters.jl
Сканирование β завершено. Результат в data/sir_scan.csv

```

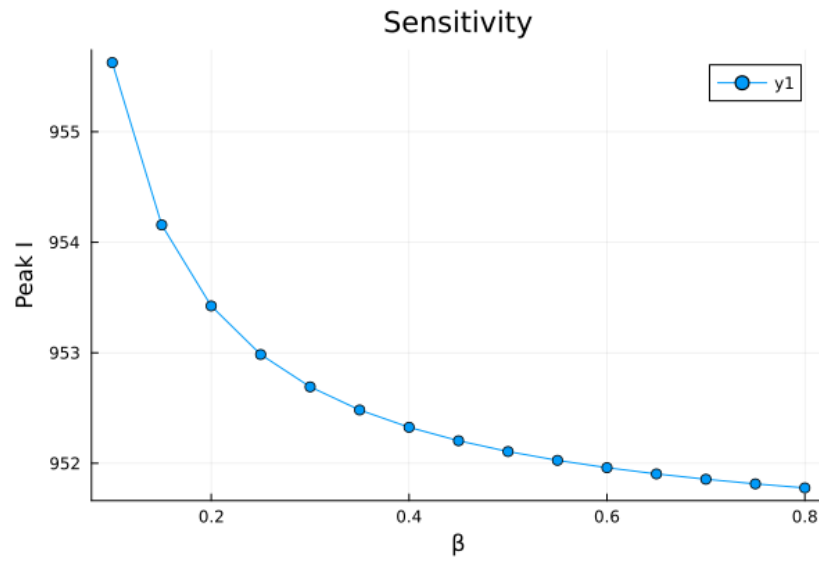
6

. I R .

5. (sirpetri_animate.jl)

GIF- , S, I, R

5



7

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab06/project$ julia --projec
t=. scripts/sirpetri_animate.jl
Сканирование  $\beta$  завершено. Результат в data/sir_scan.csv
```

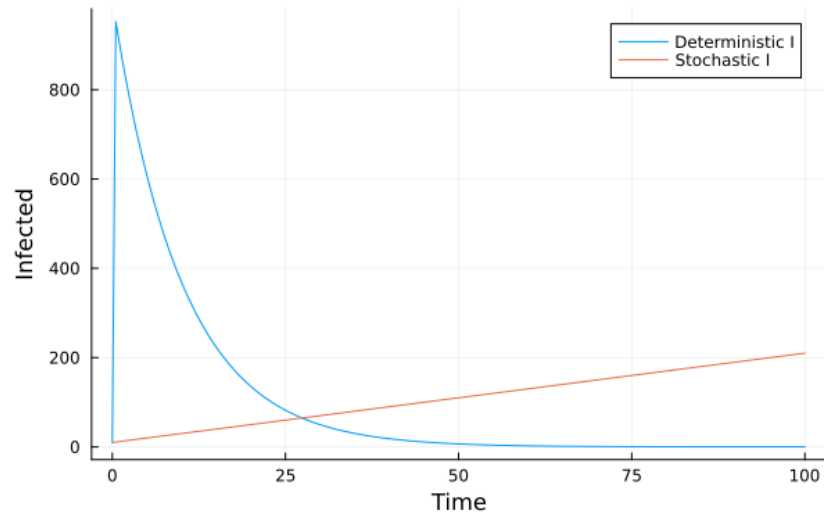
8

6

6. (sirpetri_report.jl)

: - $I(t)$ vs - I

Comparison



9

Literate : - (.jl) - Jupyter Notebook (.ipynb) - Quarto (.qmd)

Quarto

_quarto.yml Julia, .qmd-

```
execute:
  echo: true
  eval: true
```

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```

@ Base ./client.jl:552
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project$ julia --project=. scripts/tangle.jl src/SIRPetri.jl
Генерация из: src/SIRPetri.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/src/SIRPetri.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/SIRPetri/SIRPetri.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/scripts/SIRPetri/SIRPetri.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/src/SIRPetri.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/SIRPetri/SIRPetri.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/markdown/SIRPetri/SIRPetri.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/src/SIRPetri.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/SIRPetri/SIRPetri.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab06/project/notebooks/SIRPetri/SIRPetri.ipynb
Готово! Все файлы созданы.

```

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- .ipynb, .qmd

- SIR —
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- SIR Julia
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-

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- Julia +
- : ODE (Tsit5)
- :
- : Plots, Graphviz
- : Literate, Jupyter, Quarto

()

- 1.
- 2.
- 3.
- 4.
- 5.
6. Quarto

()

- S, I, R
- I
-
- Jupyter Notebook Quarto-

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